

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0924

Pine Hollow 138 kV
January 21, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0924. The Interconnection Customer proposes a 41.4 MW battery energy storage facility interconnecting at the Pine Hollow 138 kV substation in the Virginia area. The request was received on January 20, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 265 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,182,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Dekalb - Jones Chapel 500 KV
Contingency: Robison Park - Russell Fork 230 KV line
Loading: 104.1% (Rating: 4,496 MVA)
DFAX: 13.1%
Pre-project loading: 103.9%
2. Facility: Bluebonnet - Russell Fork 230 KV
Contingency: Jones Chapel - Dekalb Substation 500 KV
Loading: 103.7% (Rating: 2,119 MVA)
DFAX: 6.4%
Pre-project loading: 103.6%
3. Facility: Dekalb 230 kV/1 kV Transformer Bank
Contingency: Richland - Dekalb 500 KV
Loading: 102.8% (Rating: 2,597 MVA)
DFAX: 7.4%
Pre-project loading: 102.7%
4. Facility: Dekalb 500/1 kV Transformer
Contingency: Richland - Dekalb 500 KV
Loading: 102.8% (Rating: 2,597 MVA)
DFAX: 7.4%
Pre-project loading: 102.7%
5. Facility: Galion Generatingstation Substation - Trego 500 KV
Contingency: Beaver - Trego 500 KV
Loading: 102.7% (Rating: 3,223 MVA)
DFAX: 5.6%
Pre-project loading: 102.6%

6. Facility: Concord - Bluebonnet 230 KV
Contingency: Boca Raton - Pueblo Xtra 500 KV
Loading: 102.6% (Rating: 1,571 MVA)
DFAX: 6.2%
Pre-project loading: 102.4%
7. Facility: Imperial Substation - Eufaula 500 KV
Contingency: Trego - Galion Generating Sta 500 KV
Loading: 102.4% (Rating: 4,273 MVA)
DFAX: 8.7%
Pre-project loading: 102.3%
8. Facility: Dekalb - Robison Park 230 KV L
Contingency: Dekalb - Jones Chapel 500 KV
Loading: 100.8% (Rating: 2,827 MVA)
DFAX: 6.8%
Pre-project loading: 100.7%
9. Facility: Trego - Swan Valley 500 KV
Contingency: Trego - Troy Energy Llc 500 KV
Loading: 100.2% (Rating: 4,453 MVA)
DFAX: 5.5%
Pre-project loading: 100.2%
10. Facility: Boca Raton - Pueblo Xtra 500 KV
Contingency: Concord - Bluebonnet 230 KV
Loading: 100.2% (Rating: 2,945 MVA)
DFAX: 9.0%
Pre-project loading: 100.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dekalb - Jones Chapel 500 KV	Reconductor line	\$28,170,000	0% (\$103,000)
Bluebonnet - Russell Fork 230 KV	Reconductor line	\$21,783,000	0% (\$80,000)
Dekalb 230 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$6,023,000	0% (\$22,000)
Dekalb 500/1 kV Transformer	Replace transformer with higher-rated unit	\$25,786,000	0% (\$94,000)
Trego - Galion Generating Sta 500 KV	Reconductor line	\$66,562,000	0% (\$239,000)
Concord - Bluebonnet 230 KV	Reconductor line	\$10,631,000	0% (\$39,000)
Imperial Substation - Eufaula 500 KV	Reconductor line	\$37,039,000	0% (\$132,000)
Dekalb - Robison Park 230 KV L	Reconductor line	\$21,193,000	0% (\$78,000)
Trego - Swan Vly 500 KV line	Reconductor line	\$54,440,000	0% (\$254,000)
Boca Raton Substation - Pueblo Xtra 500 KV	Reconductor line	\$38,618,000	0% (\$141,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$1,182,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.